

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Bachelor of Technology (Mechanical Engineering) SEMESTER - 5 Winter 2025 ( Regular )**

**Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology**

**Semester : SEMESTER - 5**

**Subject Code & Name: BTAP504D - AUTOMOBILE ENGINEERING**

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**Time : 3 Hours]**

**[Total Marks : 60**

**Instructions to the Students:**

1. Each question carries 12 marks.
  2. Question No. 1 will be compulsory and include objective-type questions.
  3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
  4. Use of non-programmable scientific calculators is allowed.
  5. Assume suitable data wherever necessary and mention it clearly.
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**Q1. Objective type questions. (Compulsory Question) 12**

- 1 The main function of the automobile chassis is to: A) Reduce vehicle weight B) Increase engine power C) Support and mount all vehicle components D) Improve fuel efficiency
- 2 Which of the following is a commonly used automobile frame? A) Ladder frame B) Cross frame C) Box frame D) Tubular frame
- 3 In an inline engine, cylinders are arranged: A) In V-shape B) Opposite to each other C) In a straight line D) Radially around crankshaft
- 4 The purpose of steering geometry is to: A) Increase engine efficiency B) Avoid tyre slipping during turning C) Improve suspension comfort D) Reduce fuel consumption
- 5 Which suspension allows independent movement of wheels? A) Rigid axle suspension B) Leaf spring suspension C) Independent suspension D) Multi-axle suspension
- 6 The function of a clutch is to: A) Increase engine speed B) Engage and disengage engine power C) Reduce fuel consumption D) Change gear ratio
- 7 The advantage of a synchromesh gearbox is: A) High torque multiplication B) Smooth gear shifting C) Reduced engine size D) Lower vehicle weight
- 8 The function of a differential is to: A) Increase torque B) Reduce speed C) Allow wheels to rotate at different speeds D) Improve braking
- 9 Wheel balancing is carried out to: A) Improve fuel economy B) Reduce vibration and wobbling C) Increase tyre life only D) Improve braking power
- 10 A lead-acid battery converts: A) Mechanical energy to electrical energy B) Electrical energy to chemical energy C) Chemical energy to electrical energy D) Heat energy to electrical energy
- 11 The function of a voltage regulator is to: A) Disconnect battery B) Control battery charging rate C) Increase starter torque D) Improve ignition
- 12 In vehicle performance curves, the intersection of tractive effort and resistance curves indicates: A) Maximum acceleration B) Maximum braking force C) Maximum vehicle speed D) Minimum fuel consumption

**Q2. Solve the following.**

- A) Explain in detail the advantages and disadvantages of the following with neat sketch: 6
- Front engine rear wheel drive
  - Rear engine rear wheel drive
- B) What are vehicle performance curves? Explain speed–tractive effort and speed–power curves with suitable diagrams. 6
- Q3. Solve the following.
- A) Explain steering geometry in detail. Discuss camber, caster, toe-in, toe-out and their effects on vehicle performance. 6
- B) Describe the construction, working, and applications of the leaf spring suspension system. 6
- Q4. Solve Any Two of the following.
- A) Describe the construction and working of a synchromesh gearbox with a neat sketch. 6
- B) Classify different types of clutches used in automobiles. Explain the working of a single-plate clutch. 6
- C) Explain different types of drive systems used in automobiles. Discuss the four-wheel drive (4WD) system with its advantages. 6
- Q5. Solve Any Two of the following.
- A) Explain tyre construction and tyre specification. 6
- B) Explain the need and functions of a braking system in an automobile. Describe different types of brakes used. 6
- C) Explain the construction and working of a tubeless tyre with a neat sketch. Mention its advantages over tube-type tyres. 6
- Q6. Solve Any Two of the following.
- A) Explain emission control systems used in automobiles. Describe any One emission reduction technique. 6
- B) Explain the construction, working and maintenance of a lead-acid battery used in automobiles. 6
- C) Explain the importance of vehicle maintenance. Describe routine service procedure and troubleshooting. 6

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